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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations Oct-2020

Electronics and Communication Engineering

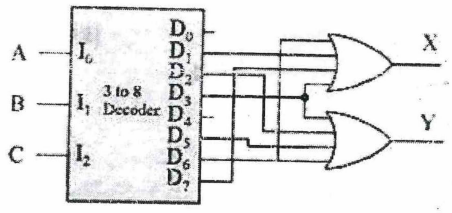
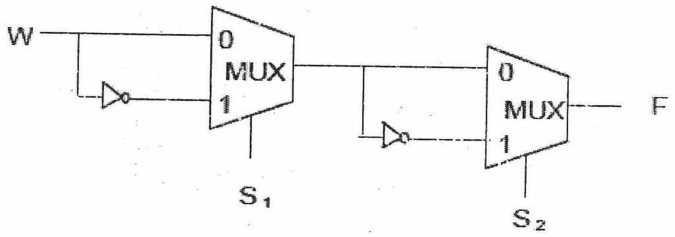
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

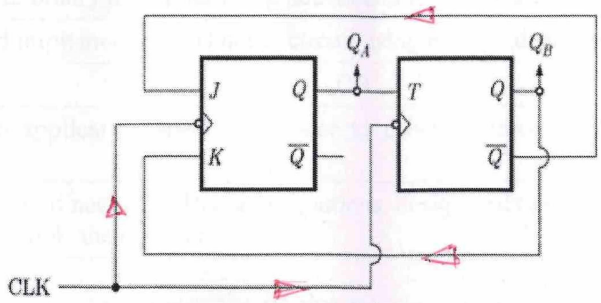
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

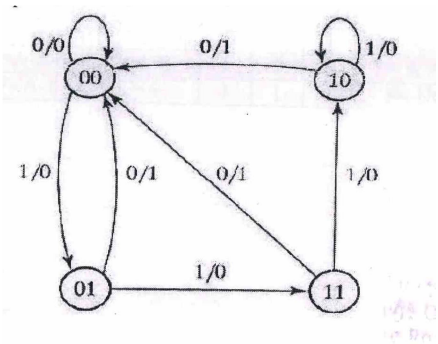
PART-A

1	1.1	The fastest digital IC logic family is _____.	01
	1.2	If the output high current of a gate is $100mA$ and its input high current is $20mA$, the fan out of the gate is _____.	01
	1.3	In the sum of products function $f = \sum m(2,3,4,5,7)$, list the essential prime implicants.	01
	1.4	In a 5-variable K-map, the cell 4 is adjacent to _____, _____, _____, _____ and _____ cells.	01
	1.5	The digital block shown in Fig1.5 is a active high output decoder. The simplified output expression for X is _____.  Fig.1.5	02
	1.6	The functionality (Y) implemented using combinatorial circuit shown in Fig 1.6 _____.  Fig.1.6	01
	1.7	Consider a two level logic implementation of the look ahead carry adder. Assume that all Pi and Gi are available for carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND and OR gates needed to implement the look ahead carry generator for 4-bit adder with $S3, S2, S1, S0$ and $C4$ as its output are _____.	01

1.8	For a $JK-FF$, if $Q(t) = 1$ and $Q(t+1) = 0$ then, the inputs at J and K are _____ respectively.	01
1.9	4-bit data 1101 is stored in a bidirectional 4-bit circular shift register. If the control input $RIGHT/\overline{LEFT}$ is high, the shift register content after three clock pulses is _____.	01
1.10	If the state Q_AQ_B of the circuit shown in Fig 1.10 at the clock time tn is '10' then, the state Q_AQ_B of the circuit at $tn+3$ (after three clock cycles) is _____.  Fig.1.10	02
1.11	4 bit ripple and synchronous counters are designed using flip-flops having propagation delay of $10ns$ each. The worst case delay in ripple and synchronous counter are _____ and _____ respectively.	01
1.12	In a Johnson counter, on the third clock pulse the sequence is 1110. On the sixth clock pulse, the sequence is _____.	01
1.13	Minimum number of flip flops required to design a sequence generator to obtain a sequence 10111 is _____.	01
1.14	The minimum number of flip flops used to design a mod-100 ripple counter is _____.	01
1.15	The processor employing accumulator register performs an operation $R1 \leftarrow R2 + R3$. The number of clock cycles required to compute the same is _____.	01
1.16	For the following set of micro operations, write the control word register contents. i) $R1 \leftarrow Crc\ R2$ ii) $R3 \leftarrow R5\ xor\ R5$	02
1.17	In a true/complement zero/one logic, if the operation "$A-1$" (decrement) is performed then the value of output y = _____.	01

PART-B

2	a	List out the basic properties of a digital logic gate. Design a two input Totem pole $TTL\ NAND$ gate and explain its working.	06
	b	Simplify the Boolean expression: $Y = \sum m(1,2,3,7,9,11,17,18,23,25,29,31) + d(5,6,10,16)$ using K-map method.	05
	c	Assume that a room has three doors and a switch by each door to control a single light in the room. Let X, Y and Z denote the state of all the switches. Assume the light is off if all the switches are open. Closing any switch turns the light ON, but closing another switch will turn the light OFF. Light is ON if all three switches are closed. Construct the truth table and implement the logic using logic gates.	05

3	a	Implement the following Boolean functions (f_1 and f_2) using suitable multiplexer. $f_1 = (y' + x)z$ and f_2 =(y'z' +x'y +yz')	06
	b	Design 3×2 binary multiplier using adders and logic gates.	05
	c	Design and implement a BCD adder circuit using binary adders suitably.	05
		OR	
4	a	Design an application specific decoder to drive common cathode seven segment display.	08
	b	With the help of necessary Boolean equations, design and explain the working of 4-bit carry look ahead adder.	08
5	a	What is race around condition in JK-FF? Design a master slave JK FF using only $NAND$ gates employing asynchronous inputs. Explain the working with the help of function table and timing diagram.	07
	b	Develop a 2-bit shift register using SR-FFs for following specifications. i) When mode=0 , the shift right operation is controlled by negative edge of $CLK1$ ii) When mode=I, load parallel data and the operation is controlled by negative edge of $CLK2$ iii) The register should contain an active low $CLEAR$.	05
	c	Construct a 4-bit Johnson counter and Draw the timing diagram for the same.	04
		OR	
6	a	Design a synchronous counter to count a sequence 0-2-4-6-0-2-4-6-0 ... using JK-FF's .	07
	b	Draw a serial adder to add two 4-bit numbers. Write the state diagram for the same and design a mechanism to stop the addition after 4th bit.	04
	c	Design a sequence generator to generate the binary sequence 10110.	05
7	a	Design a sequential circuit using D-FF's satisfying following state diagram given in Fig 7a with the help of state table.  Fig.7(a)	08
	b	Design a mod-6 ripple up counter using T-Flip-Flops.	04

c For the sequential logic represented by the state diagram shown in Fig 7c, perform state reduction and draw the reduced state diagram.

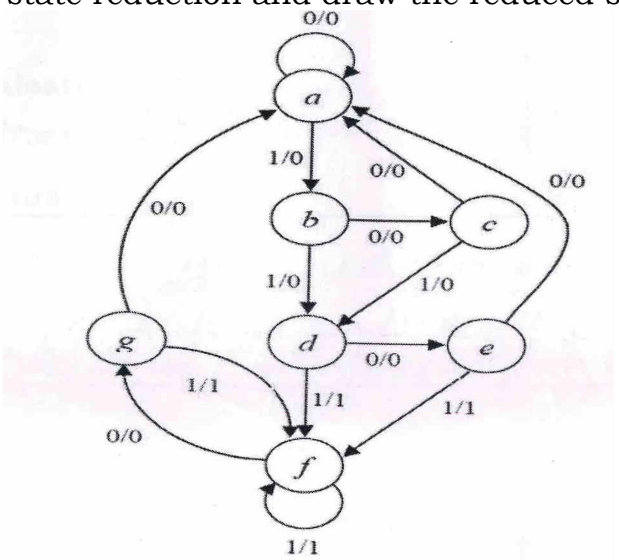


Fig.7(c) 04

8 a What is a scratch pad memory? Explain the organization of a processor unit using scratch pad memory with the help of a neat diagram. 04

b With a neat diagram, explain the status register of a processor. 04

c Design an arithmetic and logic circuit, with three selection variables S_2, S_1 and S_0, that generates the following operation. Draw the logic diagram of one typical stage.

Arithmetic Operations:					*Logical operations*:			
S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	S_2	S_1	S_0	$C_{in} = X$
0	0	0	$F = A$	$F = A + 1$	1	0	0	$F = A\ OR\ B$
0	0	1	$F = A + B$	$F = A + B + 1$	1	0	1	$F = A\ XOR\ B$
0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	$F = A\ AND\ B$
0	1	1	$F = A - 1$	$F = A$	1	1	1	$F = NOT\ A$

08